

LoadRunner Interview questions and answers for basic to advance

Basic Level – Fundamentals

1. What is LoadRunner?

Answer:

LoadRunner is a performance testing tool by Micro Focus used to test the behavior and performance of applications under load. It simulates multiple users to measure response times, throughput, and bottlenecks.

2. What are the components of LoadRunner?

Answer:

1. **VUGen (Virtual User Generator)** – Records and creates scripts.
 2. **Controller** – Designs and executes load tests.
 3. **Load Generator** – Generates virtual user load.
 4. **Analysis** – Analyzes results after execution.
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3. What is a Vuser?

Answer:

A **Vuser (Virtual User)** is a simulated user that performs operations on the application as a real user would.

4. What is Correlation in LoadRunner?

Answer:

Correlation is used to handle dynamic values (like session IDs, tokens) that change with each request.

- **Manual correlation:** Using `web_reg_save_param()`
 - **Automatic correlation:** Through rules during recording.
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5. What is Parameterization in LoadRunner?

Answer:

Parameterization replaces hard-coded input values (like username, password) with variables to simulate real user data and avoid server-side caching.

6. What is Think Time?

Answer:

The delay between user actions to simulate real human behavior.
Use `lr_think_time(seconds);`

7. What is Pacing?

Answer:

Pacing controls how often each Vuser iteration runs.

Example:

$$\text{Pacing} = \frac{3600 \text{ iterations/hour}}{\text{Response Time} + \text{Think Time}}$$

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8. What are the types of Scenarios in LoadRunner?

Answer:

1. **Manual Scenario** – You manually assign scripts and users.
 2. **Goal-Oriented Scenario** – LoadRunner automatically adjusts users to meet a performance goal (e.g., 100 txn/sec).
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9. What are common protocols supported by LoadRunner?

Answer:

HTTP/HTML, Web Services, TruClient, Citrix, SAP, Oracle, Java, .NET, and more.

10. What are Transactions in LoadRunner?

Answer:

Transactions measure the time between specific start and end points in a script using:

```
lr_start_transaction("Login");  
... steps ...  
lr_end_transaction("Login", LR_AUTO);
```

Intermediate Level – Real-Time Concepts

11. What is Rendezvous Point?

Answer:

It synchronizes multiple users to perform a task simultaneously — helps simulate peak load.

```
lr_rendezvous("CheckoutPoint");
```

12. What is the difference between Throughput and Hits per second?

Answer:

- **Throughput** – Amount of data sent/received per second (KB/sec).
 - **Hits/sec** – Number of HTTP requests sent per second.
More hits ≠ better performance — depends on server response capacity.
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13. What is a Bottleneck?

Answer:

A bottleneck is a point in the system where performance is limited (e.g., CPU, memory, DB query, or network latency).

14. How do you identify bottlenecks in LoadRunner?

Answer:

- Compare **response time**, **throughput**, and **hits/sec** graphs.
 - Monitor **server-side metrics** (CPU, memory, DB).
 - Check **transaction breakdown** and **error count**.
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15. What is Error Handling in LoadRunner?

Answer:

Use `lr_error_message("Error found...");` and handle expected failures using `continue on error` in Runtime Settings.

16. What is the purpose of the Ramp-up and Ramp-down period?

Answer:

- **Ramp-up:** Gradually increase users to avoid sudden load.
 - **Ramp-down:** Gradually decrease load after steady state.
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17. What is Think Time and Pacing difference?

Think Time	Pacing
Simulates user delay	Controls iteration frequency
Between steps	Between iterations

18. Explain Little's Law in Performance Testing.

Answer:

$$N = X \times R \quad N = X \times R \quad N = X \times R$$

Where:

- N = Number of users
 - X = Throughput (transactions/sec)
 - R = Response Time
- Helps validate test consistency and steady state.
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19. How do you calculate throughput from test results?

Answer:

$$\text{Throughput} = \frac{\text{Total data transferred}}{\text{Total time taken}} \quad \text{Throughput} = \frac{\text{Total data transferred}}{\text{Total time taken}}$$

20. What are common graphs analyzed in LoadRunner Analysis?

- Transaction Response Time graph
- Running Vusers graph
- Throughput graph
- Hits/sec
- Error count graph
- Network delay and breakdown graphs

Advanced Level – Real Time Scenarios

21. How do you design a workload model?

Answer:

1. Identify business transactions.
2. Determine user distribution (% per transaction).
3. Calculate concurrency (active users).
4. Set pacing & think time.
5. Create test mix in Controller.

22. How do you calculate the number of users for a test?

Answer:

Use **Little's Law** or **NFR (Non-Functional Requirements)**:

$$\text{Users} = \text{TPS} \times \text{Average Response Time}$$
$$\text{Time Users} = \text{TPS} \times \text{Average Response Time}$$

23. Explain different types of Performance Tests.

- **Load Test** – Expected load.
- **Stress Test** – Beyond expected load.
- **Endurance/Soak Test** – Long duration.
- **Spike Test** – Sudden load increase.
- **Scalability Test** – Verify system growth capacity.

24. What is an SLA (Service Level Agreement)?

Answer:

It's a performance target agreed upon:

- Response time < 3 sec
- Error rate < 1%
- CPU utilization < 80%

25. What is the difference between LoadRunner Web (HTTP/HTML) and TruClient protocol?

Web (HTTP/HTML)	TruClient
Protocol-level recording	Browser-level recording
Lightweight & faster	Heavy but realistic
Ideal for APIs/web	Ideal for UI-based apps

26. What is the significance of 'Think Time ignored' in Runtime Settings?

Answer:

Used in **benchmark testing** or **stress tests** where user delay is not simulated.

27. How do you correlate dynamic session IDs manually?

Answer:

1. Record twice.
2. Identify changing value.
3. Use `web_reg_save_param()` before the request.

Example:

```
web_reg_save_param("SessionID", "LB=sessionID=", "RB=;", LAST);
```

28. What are the different types of Logs in LoadRunner?

- **Standard Log** – Logs API and message outputs.
 - **Extended Log** – Includes parameter and server data.
 - **Disable Logging** – Used in execution to reduce overhead.
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29. What is a Checkpoint in LoadRunner?

Answer:

Used to **verify expected content** on the response page using `web_find()` or `text_check()`.

30. How do you validate the performance test results?

Answer:

1. Check steady-state period.
2. Verify SLAs.
3. Correlate response time vs throughput.
4. Review server metrics for bottlenecks.
5. Document deviations.

31. What are best practices in LoadRunner scripting?

- Use correlation and parameterization properly.
 - Avoid hard-coded values.
 - Disable logging during execution.
 - Use minimal think times.
 - Handle dynamic data with regular expressions.
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32. What is the difference between Transaction Response Time and Page Load Time?

Transaction RT	Page Load Time
Defined in script manually	Measured at browser end
Measures logical flow	Measures full render time

33. What is the difference between Hits/sec and Transactions/sec?

Hits/sec	Transactions/sec
Number of HTTP requests	Number of business transactions
Low-level metric	High-level user metric

34. How do you perform parameter correlation using JSON Extractor or Regular Expression in LoadRunner?

Use `web_reg_save_param_json()` for JSON-based responses or regex rules for text.

35. How to handle SSL certificate errors in LoadRunner?

Set SSL versions under **Runtime Settings** → **Internet Protocol** → **Preferences** → **Advanced**.

Advanced LoadRunner Interview Questions (with Answers)

36. How do you determine the steady state in a performance test?

Answer:

Steady state is when the system performance metrics (response time, throughput, and CPU/memory usage) remain consistent for a continuous period (usually 10–15 minutes). You can verify it in the **Analysis graphs**:

- Response time curve stabilizes.
 - Throughput remains constant.
 - No sudden spikes or drops.
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37. How do you calculate Transactions Per Second (TPS)?

Answer:

$$\text{TPS} = \frac{\text{Total successful transactions}}{\text{Test duration (in seconds)}}$$
$$\text{TPS} = \frac{\text{Total successful transactions}}{\text{Test duration (in seconds)}}$$

Example:

10,000 transactions in a 10-minute test →

$$\text{TPS} = 10000 / 600 = 16.6$$
$$\text{TPS} = 10000 / 600 = 16.6$$

38. What is the difference between Load, Stress, and Endurance testing?

Type	Description	Purpose
Load	Test under expected load	Validate SLAs
Stress	Test beyond expected load	Find system break point
Endurance (Soak)	Run for long duration	Check memory leaks & stability

39. What is “Controller Scheduling” in LoadRunner?

Answer:

It defines **when and how** virtual users start and stop.

You can configure:

- **Ramp-up** (e.g., 10 users every 30 sec)
 - **Steady state duration** (e.g., 30 min)
 - **Ramp-down** (e.g., stop 10 users every 30 sec)
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40. What is the use of the “Runtime Settings” window?

Answer:

It controls script behavior during execution:

- Think Time
- Pacing

- Logging level
 - Error handling
 - Proxy settings
 - Automatic transactions
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41. What is “Script Check-In/Check-Out” in LoadRunner Enterprise (Performance Center)?

Answer:

It allows version control. Only one tester can check out and modify a script at a time; others can only view until it's checked back in.

42. What are common causes of high response time in LoadRunner tests?

- Network latency
 - Database query delays
 - CPU/memory bottlenecks on servers
 - Thread locking/contention
 - High page/component size
 - Unoptimized code
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43. How do you validate server-side bottlenecks during load testing?

Answer:

Use **LoadRunner monitors** or external tools:

- **Web Server Monitor:** CPU, threads, queue length
 - **Database Monitor:** DB locks, query time
 - **System Resource Monitor:** CPU, memory, disk I/O
 - **Network Monitor:** Bandwidth, latency
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44. How do you correlate dynamic values in JSON response?

Answer:

```
web_reg_save_param_json(  
    "ParamName=AuthToken",  
    "QueryString=$.token",  
    LAST);
```

This extracts the value of token from the JSON response.

45. What is the use of the lr_rendezvous() function?

Answer:

To emulate peak load by making multiple users perform the same operation simultaneously:

```
lr_rendezvous("LoginPeakLoad");
```

46. What are common performance bottlenecks identified during LoadRunner tests?

- Web server saturation
 - Database connection pool limits
 - Memory leaks in application server
 - Slow SQL queries
 - Thread pool exhaustion
 - Insufficient network bandwidth
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47. How do you handle dynamic tokens that appear only after login?

Answer:

Use **correlation**:

1. Record login.
 2. Identify token (e.g., SessionID).
 3. Add web_reg_save_param() **before** login request.
 4. Use captured token in subsequent requests.
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48. What are the different types of monitors in LoadRunner?

- **Server Monitors** (CPU, memory, disk, etc.)
 - **Application Monitors** (WebLogic, WebSphere, JBoss)
 - **Database Monitors** (Oracle, SQL Server)
 - **Network Monitors** (LAN/WAN)
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49. What are the best practices for LoadRunner scripting?

- Always use correlation for dynamic data.
- Parameterize all unique inputs.
- Validate key transactions.
- Use minimum logging.

- Insert proper transactions (lr_start_transaction()).
- Set pacing and think time to match real behavior.

50. What is an Error % in LoadRunner results, and what is acceptable?

Answer:

Error % = (Failed Transactions / Total Transactions) × 100

Acceptable limit = < 2% (depends on SLA).

51. What is the difference between Average and Percentile Response Time?

Metric	Meaning
Average	Mean of all response times
90th Percentile	90% of transactions completed within this time (used in SLAs)

52. How do you use Think Time dynamically?

Answer:

Use randomization:

lr_think_time(rand() % 5 + 2); // between 2–7 seconds

53. What are the benefits of using LoadRunner Analysis tool?

- Auto-correlation between graphs
- SLA comparison
- Throughput vs response time validation
- Error detection
- Report generation (Word/HTML/PDF)

54. What is the difference between LoadRunner and JMeter?

Feature	LoadRunner	JMeter
Protocols	Wide range	Mostly web/app
Cost	Licensed	Open source
UI	Advanced GUI	Simple
Monitoring	Built-in	Limited / via plugins
Ideal for	Large enterprise	Small-medium scale

55. What is a Workload Model?

Answer:

A representation of how users interact with the system in real life — includes:

- User distribution by business process
 - Concurrent users
 - Pacing and think time
 - Test duration
 - Transaction mix
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56. Explain correlation vs parameterization.

Correlation	Parameterization
Handles dynamic server-generated values	Handles static input data (usernames, IDs)
Done after recording	Done before execution
Prevents script failure	Simulates real users

57. What's the difference between ramp-up and ramp-down in LoadRunner?

- **Ramp-up:** Gradually increases users to reach target load.
 - **Ramp-down:** Gradually decreases users to end the test gracefully.
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58. What are "Transactions" and why are they important?

Answer:

Transactions measure response time for critical actions, e.g., login, search, checkout. They help evaluate SLA compliance.

59. What are "Iteration" and "Duration" modes in Controller?

- **Iteration mode:** Run each user for a set number of iterations.
 - **Duration mode:** Run test for a specific time (e.g., 1 hour steady state).
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60. What is a Real-Time Analysis Correlation Example?

If **response time increases** while **throughput decreases**, it usually means:

- Server resource limit reached
- Queue buildup started
- Bottleneck (CPU/memory/DB)

61. Explain the Performance Testing Life Cycle (PTLC) you follow in LoadRunner projects.

Answer:

- 1. NFR Gathering:**
 - Identify SLAs (Response time, throughput, users, TPS, error rate).
 - Example: *"Login API should respond < 2s for 500 concurrent users."*
- 2. Workload Modeling:**
 - Identify business flows, user mix, pacing, and think time.
 - Example: *60% Search, 30% Add-to-cart, 10% Checkout.*
- 3. Scripting (VUGen):**
 - Record & customize scripts (correlation, parameterization, error handling).
- 4. Scenario Design (Controller):**
 - Configure ramp-up, steady state, pacing, and think time.
- 5. Test Execution:**
 - Run multiple cycles — baseline, load, stress, endurance.
- 6. Result Analysis (Analysis Tool):**
 - Check throughput, response time, errors, steady state.
- 7. Bottleneck Identification & Tuning:**
 - Coordinate with dev/infra teams for DB, CPU, memory issues.

62. How do you gather NFRs (Non-Functional Requirements) for a LoadRunner project?

Answer:

You gather NFRs from:

- Business analysts or architects.
- Previous production usage reports.
- Monitoring data (AppDynamics, Dynatrace, etc.).

Typical NFR parameters:

NFR Type	Example
Response Time	< 3 sec for 90th percentile
Concurrent Users	500 users
Throughput	100 TPS
Error Rate	< 1%
CPU	< 80%
Memory	< 75%

63. What is the difference between Peak Load and Average Load?

Term	Meaning
Average Load	Normal day-to-day user volume
Peak Load	Highest expected concurrent users (e.g., sale season)

Example:

If daily avg = 300 users, peak = 900 users during sale → design separate test for peak.

64. Explain a real-time workload model example.

Example:

Business Scenario	% of Users	Iterations per Hour	Think Time (sec)	Pacing (sec)
Login	10%	10	5	30
Search Product	50%	30	3	20
Add to Cart	25%	15	4	25
Checkout	15%	10	5	40

➡ This ensures realistic mix of user behavior under load.

65. What challenges did you face while scripting in LoadRunner?

Answer:

- Correlating complex dynamic values (multi-level JSON/XML).
 - Handling encrypted tokens or cookies.
 - Dealing with AJAX and dynamic requests.
 - Managing SSL certificates.
 - Parameterizing data from databases.
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66. How do you validate a LoadRunner script?

Answer:

- Run in **Replay mode** (single user).
- Check for **success responses (HTTP 200/OK)**.
- Validate **expected content** with `web_reg_find()`.
- Ensure **correlations** work dynamically.
- Disable **think time** and **pacing** temporarily during debugging.

67. How do you handle authentication tokens in LoadRunner?

Answer:

1. Capture token using:

```
web_reg_save_param_json("ParamName=AuthToken", "QueryString=$.token", LAST);
```

2. Use it in headers:

```
"Authorization=Bearer {AuthToken}",
```

68. What are the performance counters you monitor during testing?

Answer:

Application Server:

- CPU Utilization, Thread count, GC time.

Web Server:

- Requests/sec, Queue length, Network usage.

DB Server:

- Active sessions, Query response time, I/O waits.

Load Generator:

- CPU < 80%, no memory leaks.
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69. How do you verify server saturation?

Answer:

If CPU utilization hits 100% and response time increases, while throughput drops — system is **CPU-bound**.

Similarly:

- High disk I/O → Disk bottleneck.
 - High DB waits → DB bottleneck.
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70. Explain a real-time bottleneck analysis example.

Scenario: During 1000-user test, avg response time increased from 2s → 10s, throughput dropped by 40%.

Investigation:

- Web CPU = 50%, App CPU = 95%, DB CPU = 60%.
 ➡ Root cause: **App server thread pool exhausted**.
 Fix: Increase thread pool size, retest → response time stabilized to 3s.
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71. What are the main graphs you analyze post-test?

Graph Name	Purpose
Response Time vs Throughput	Identify bottlenecks
Running Vusers	Check load ramp-up
Transaction Summary	SLA validation
Errors Graph	Identify failed transactions
CPU/Memory Graphs	Check resource utilization

72. How do you ensure test reliability?

- Maintain steady-state period.
 - Run multiple iterations and compare results.
 - Ensure network stability.
 - Validate load generators are healthy.
 - Avoid background processes.
-

73. What is correlation function for JSON in LoadRunner?

```
web_reg_save_param_json(  
    "ParamName=SessionID",  
    "QueryString=$.session.id",  
    LAST);
```

This extracts nested JSON parameters dynamically.

74. What is the role of Think Time and Pacing in achieving desired throughput?

Answer:

They control **user behavior frequency**.

Throughput (TPS) depends on:

$$\text{TPS} = \frac{\text{Users} \times \text{Iterations}}{\text{Test Duration}}$$
$$\text{TPS} = \frac{\text{Test Duration}}{\text{Users} \times \text{Iterations}}$$

Adjust pacing to match TPS goals (as per NFR).

75. How to validate Little's Law ($N = X \times R$) using test data?**Example:**

- Active users (N) = 500
- Avg Response Time (R) = 5 sec

$$X = N/R = 500/5 = 100 \text{ TPS}$$
$$X = N / R = 500 / 5 = 100 \text{ TPS}$$
$$X = N/R = 500/5 = 100 \text{ TPS}$$

If controller shows ~100 TPS → test is consistent (steady state).

76. Explain a real-time performance issue you handled.**Answer Example:**

During a 1000-user load test, checkout API latency increased to 12s.

Analysis showed DB CPU at 90%, query "get_cart_items" taking 8s.

After adding index on cart table → response improved to 2.5s, TPS improved 60%.

77. How do you report performance results?**Answer:**

Report includes:

- Test Summary
 - Environment details
 - Workload model
 - Metrics (TPS, Response time, Errors)
 - Graphs comparison (Baseline vs Load)
 - Observations + Recommendations
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78. What are common reasons for failed transactions?

- Incorrect correlation
 - Network timeout
 - Dynamic value mismatch
 - Authentication failure
 - Backend service down
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79. What is the difference between Controller and Performance Center (LoadRunner Enterprise)?

Controller	Performance Center
Standalone desktop tool	Web-based centralized tool
Local execution	Distributed test management
Single user	Multi-user collaboration
Manual scheduling	Centralized scheduling/reporting

80. How do you ensure realistic load simulation?

Answer:

- Include appropriate **think time** and **pacing**.
 - Use **realistic data** (parameterization).
 - Randomize workflows.
 - Use **network virtualization** if needed.
 - Synchronize users for peaks using `lr_rendezvous()`.
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81. What kind of tuning recommendations do you provide post-test?

Examples:

- Optimize SQL queries & indexes.
 - Enable server-side caching.
 - Add load balancer or more app nodes.
 - Tune JVM heap, GC settings.
 - Increase thread pool size.
 - Optimize image/compression on web server.
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82. How do you simulate 10,000 users if license supports only 1000?

Answer:

- Run test in **phases (batches)**.
 - Use **IP spoofing** or **multiple LGs**.
 - Use **Performance Center (Enterprise)** for distributed execution.
 - Extrapolate metrics for estimation.
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83. How do you debug script replay failures?

- Compare replay vs recording logs.
 - Enable extended logs.
 - Check correlation placement.
 - Validate dynamic data and headers.
 - Use “Snapshots” to compare responses.
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84. What’s your approach to analyzing large test results?

- Identify test duration, ramp-up/down, steady-state.
- Focus on 90th percentile response time.
- Filter graphs to steady-state window.
- Correlate response time with resource utilization.
- Highlight bottlenecks and improvement points.